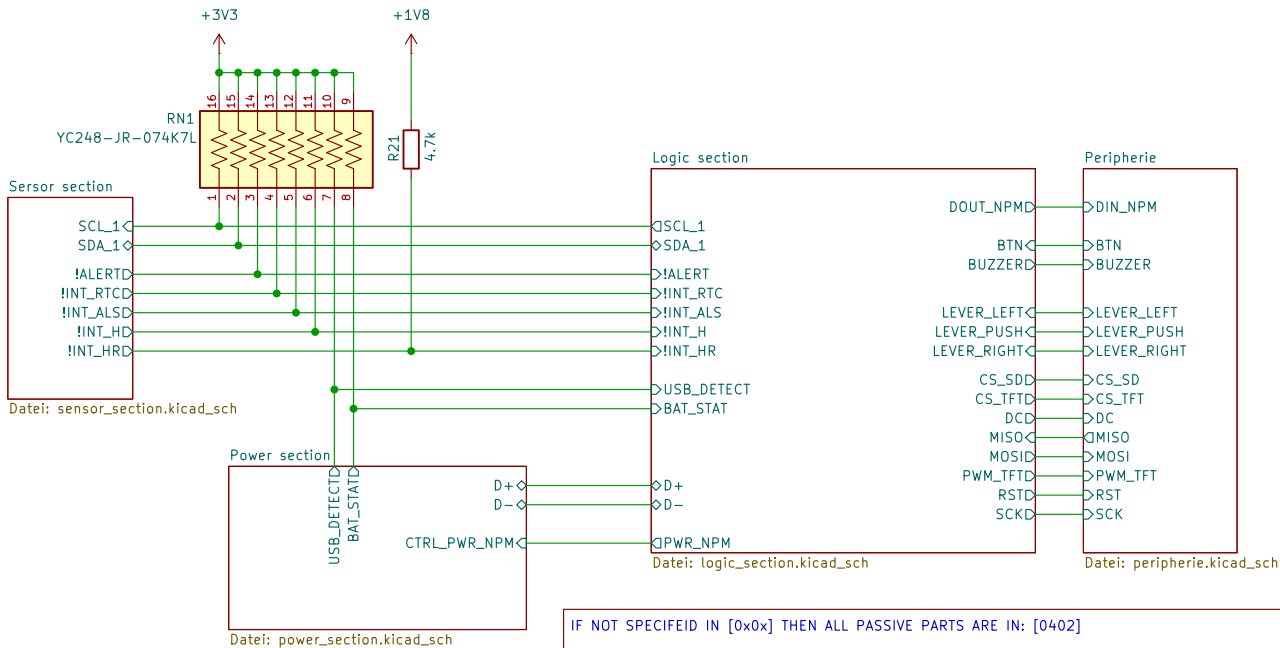


Pull-Ups



IF NOT SPECIFEID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /
File: NX-ISD.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.5

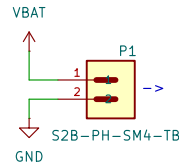
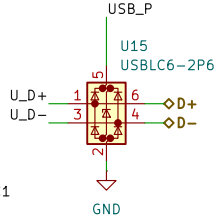
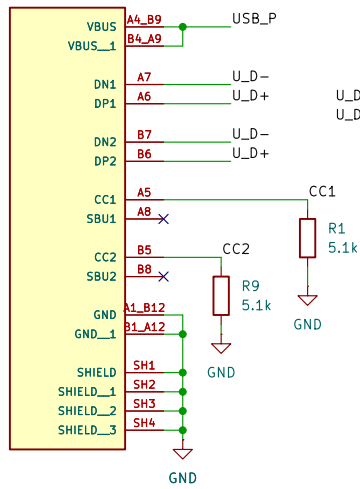
Date:

Rev:

Id: 1/5

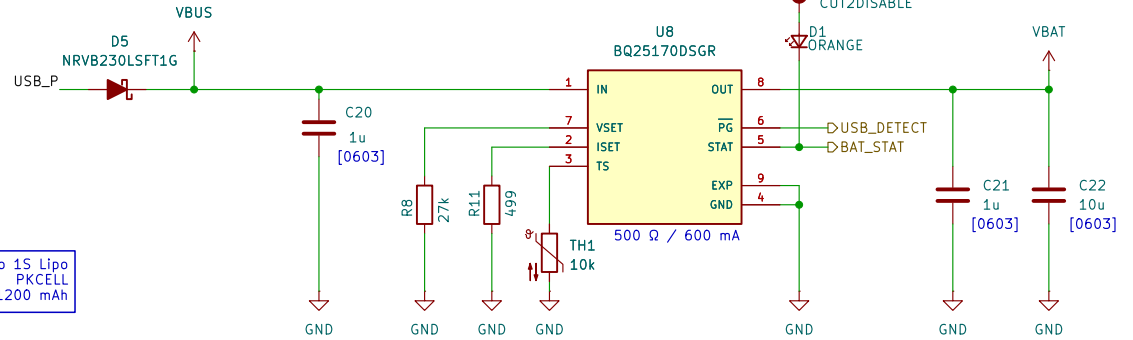
CONNECTORS

J2
UJ20-C-H-G-SMT-1-P16-TR



Connects to 1S Lipo
PKCELL
3.7 V, 1200 mAh

LIPO CHARGER

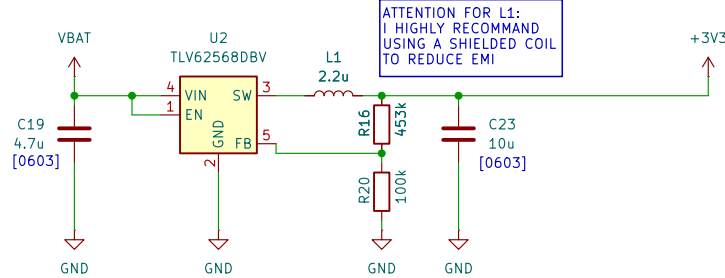


Formula $R_{ISET} = 300 / I_{OUT}(\text{in Amperes})$

Charge Current	Calculated R_{ISET}
50 mA	6000 Ω
100 mA	3000 Ω
150 mA	2000 Ω
200 mA	1500 Ω
300 mA	1000 Ω
400 mA	750 Ω
500 mA	600 Ω
600 mA	500 Ω
800 mA	375 Ω

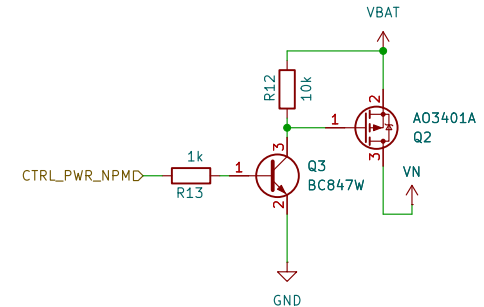
+3V3 DCDC Buck

DCDC: 3.3V / 1A



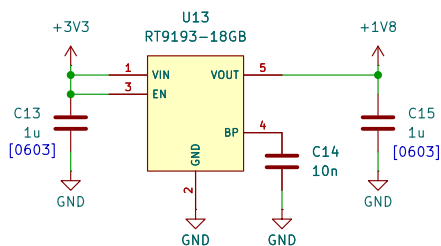
ATTENTION FOR L1:
I HIGHLY RECOMMEND
USING A SHIELDED COIL
TO REDUCE EMI

NPM POWER CONTROL



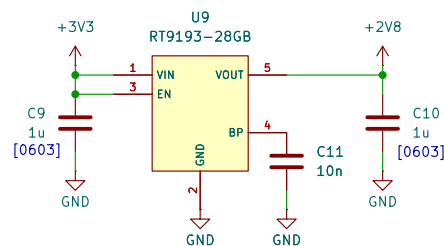
+1V8 LDO

(USED FOR HRM)



+2V8 LDO

(USED FOR DISPLAY)



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Power section/
File: power_section.kicad_sch

Title:

Size: A4

Date:

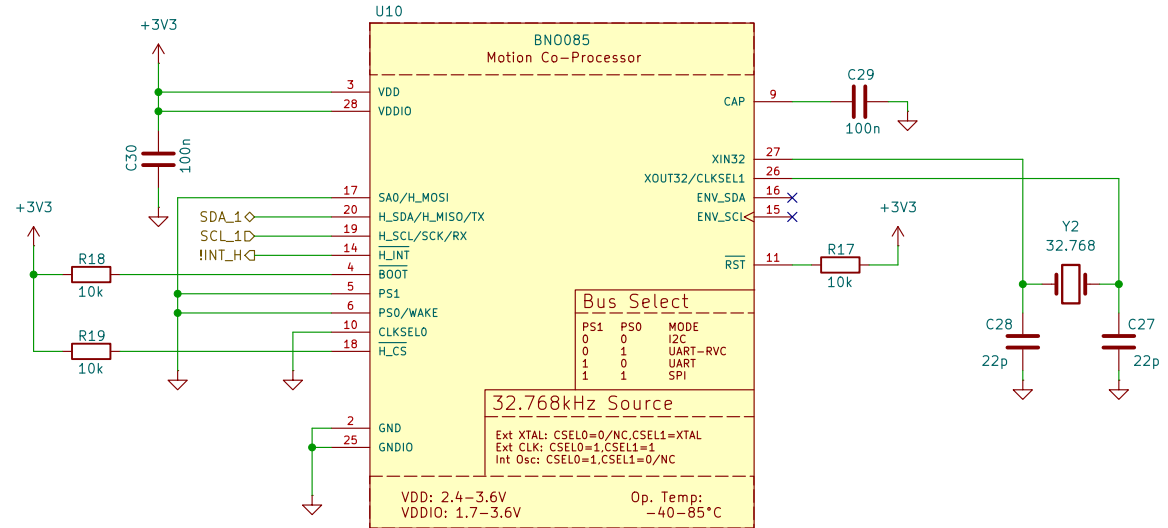
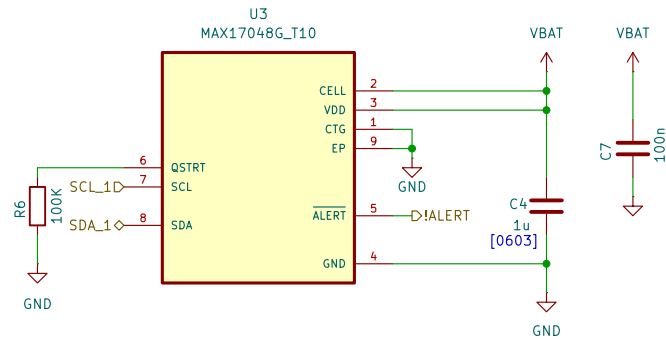
KiCad E.D.A. 10.0.5

Rev:

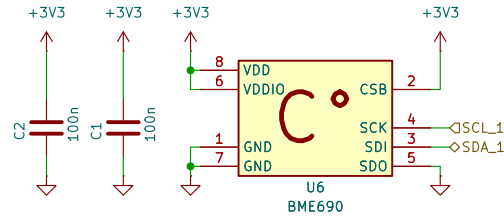
Id: 2/5

9-DOF

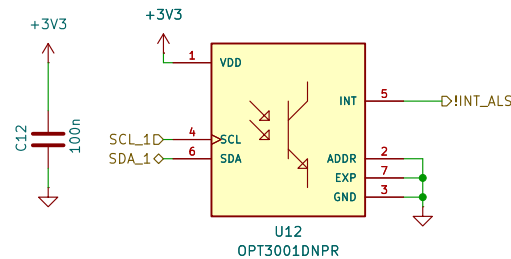
Battery Fuel Gauge



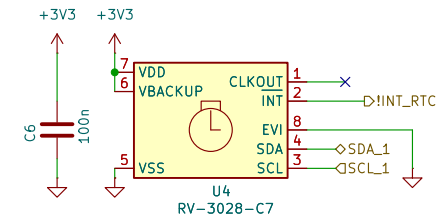
Temp/Humidity/Pressure



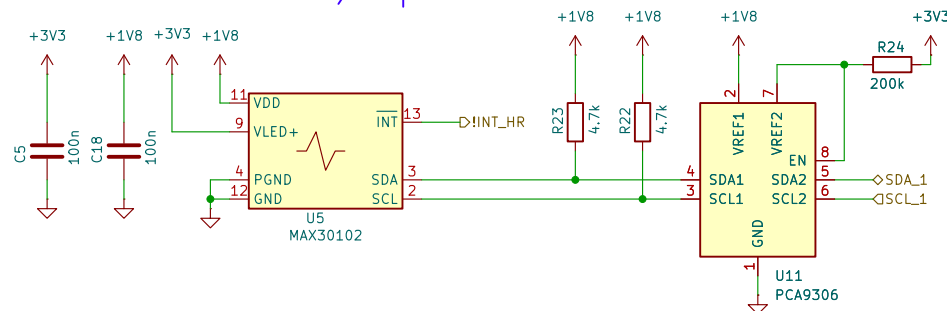
Ambient Light Sensor



RTC



BPM/SpO2



IF NOT SPECIFD IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Sensor section/
File: sensor_section.kicad_sch

Title:

Size: A4

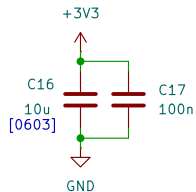
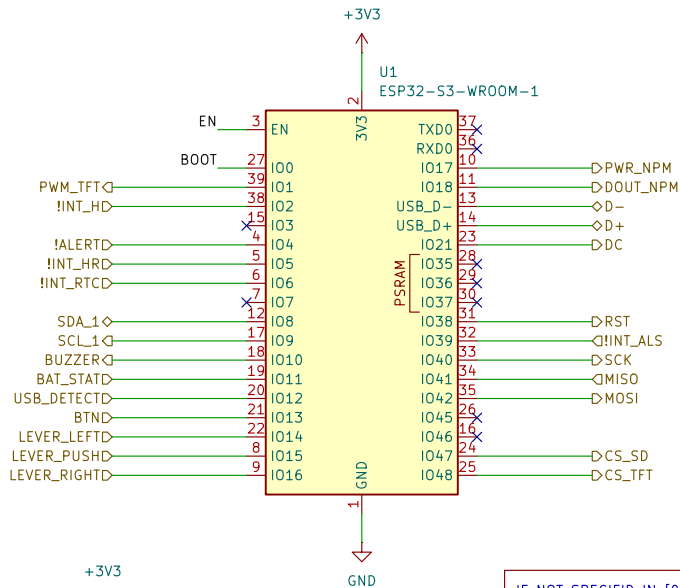
Date:

KiCad E.D.A. 10.0.5

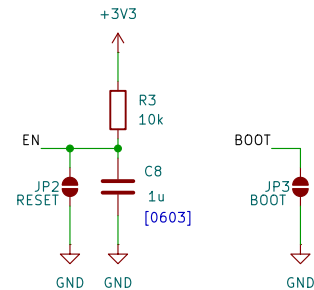
Rev:

Id: 3/5

ESP32-S3 N16R8



EN & BOOT



ESP32-S3 chip strapping pins:

GPIO 0.
GPIO 3.
GPIO 45.
GPIO 46.

IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Logic section/
File: logic_section.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.5

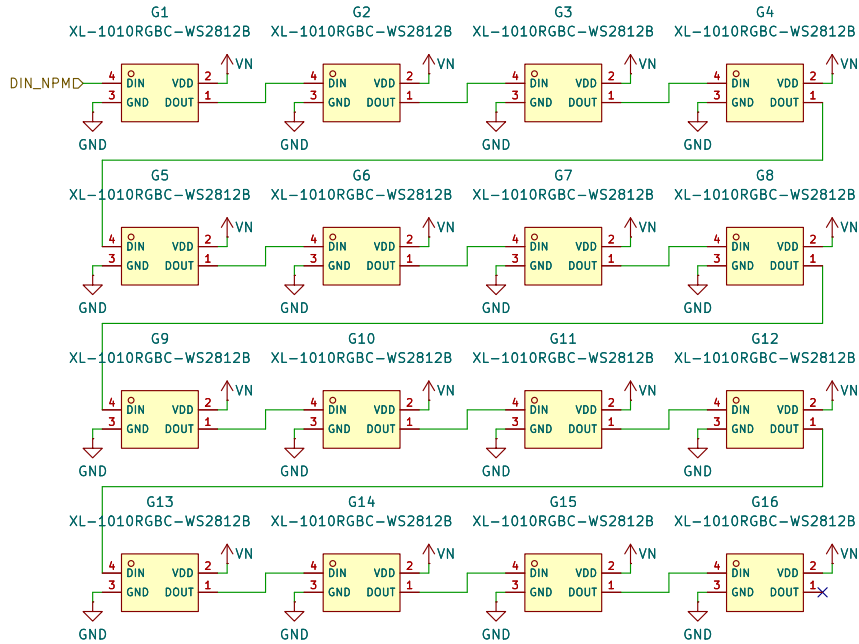
Date:

Rev:

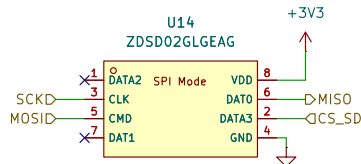
Id: 4/5

NeoPixel Matrix (NPM)

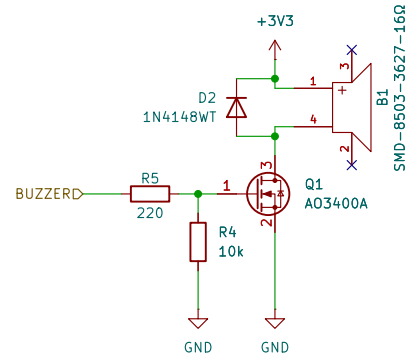
(NPM DOES NOT REQUIRE DECOUPLING CAPACITORS DUE TO THE SHORT PATH OF ITS SUPPLY)



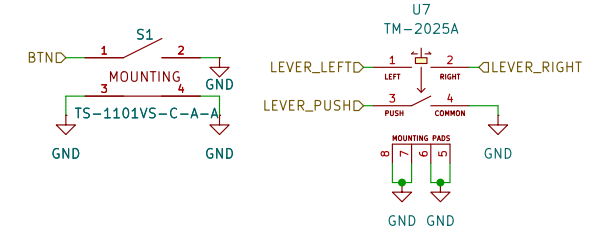
NAND-SD



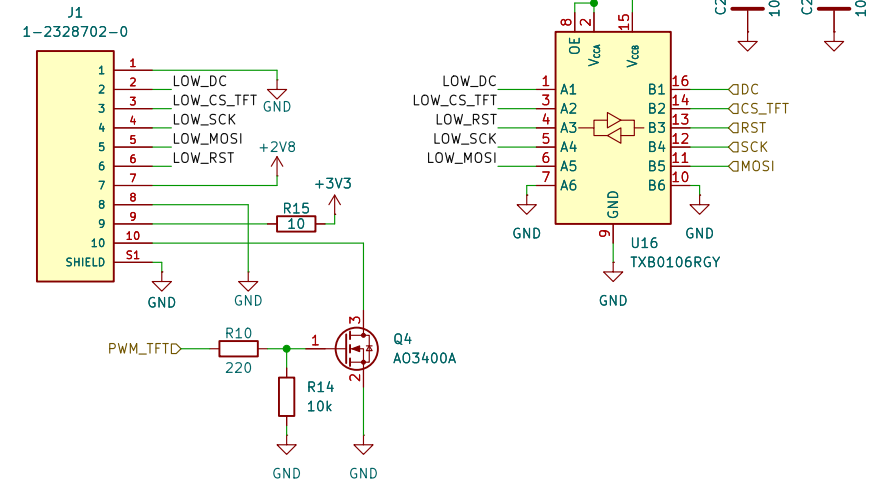
BUZZER



BUTTONS



DISPLAY



Sheet: /Peripherie/
File: peripherie.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.5

Rev:

Id: 5/5